# CARBON FOOTPRINT DASHBOARD

Tracking $CO_2$ emissions across activities, locations and time | Executive overview

Executive Dashboard | Emission Analysis | Sustainability Analysis | ML Insights

Year: Year

Location: Location

Category: Category

Activity: Activity

Total CO2
15.57K

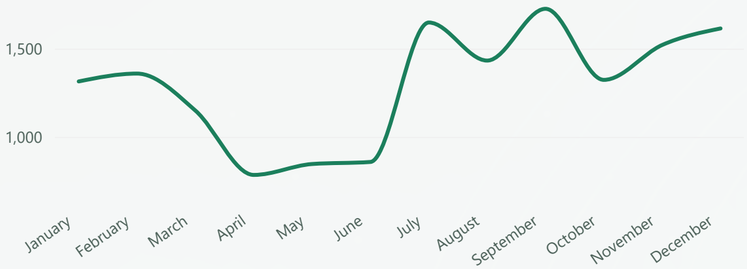
Average CO2
86.51

Maximum CO2
406.72

Number of Activities
180

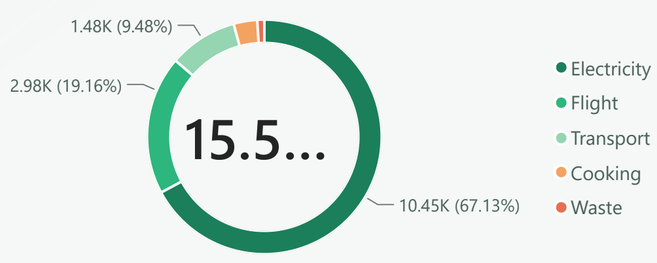
## Monthly $CO_2$ Emission Trend

Total CO2 by Month



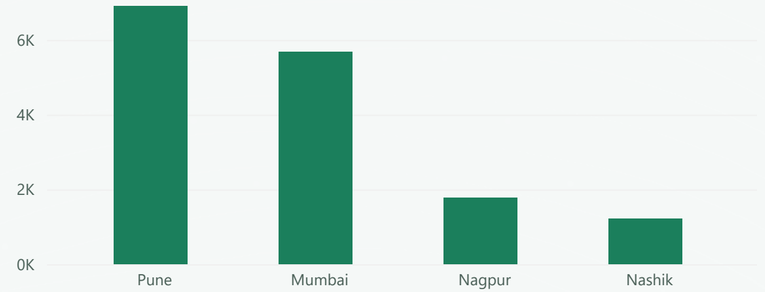

## Emissions by Category

Total CO2 by Category



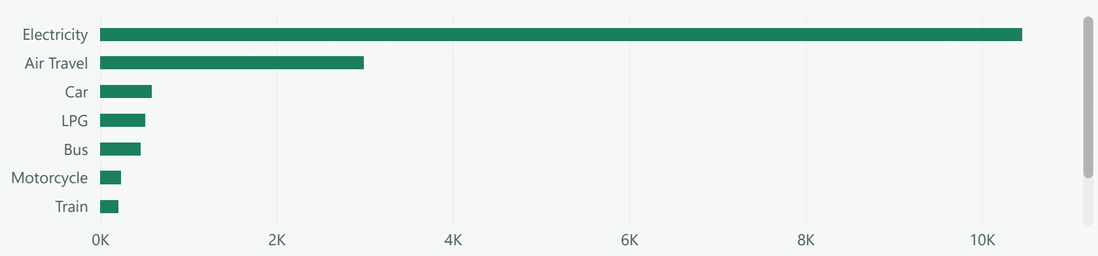

## Emissions by Location

Total CO2 by Location




## Top Emitting Activities

Total CO2 by Activity






## Path to a lower footprint

- Cut transport emissions: carpool, rail, EVs
- Switch to renewable & efficient energy
- Reduce, reuse and recycle waste
- Track monthly and set a $CO_2$ target

Use the filters above to explore every chart.

# EMISSION ANALYSIS

Where do emissions come from? Category × location breakdown, mix and activity drivers

Year

Year

## Heatmap: $CO_2$ by Category and Location

| Category | Mumbai | Nagpur | Nashik | Pune | **Total** |
|---|---|---|---|---|---|
| Cooking | 119.40 | 88.80 | 38.70 | 259.80 | **506.70** |
| Electricity | 3,803.98 | 1,102.08 | 875.76 | 4,671.54 | **10,453.36** |
| Flight | 1,274.85 | 312.90 | 87.90 | 1,308.45 | **2,984.10** |
| Transport | 389.82 | 259.72 | 194.32 | 632.09 | **1,475.95** |
| Waste | 82.70 | 15.87 | 14.75 | 38.56 | **151.88** |
| **Total** | **5,670.75** | **1,779.37** | **1,211.43** | **6,910.44** | **15,571.99** |

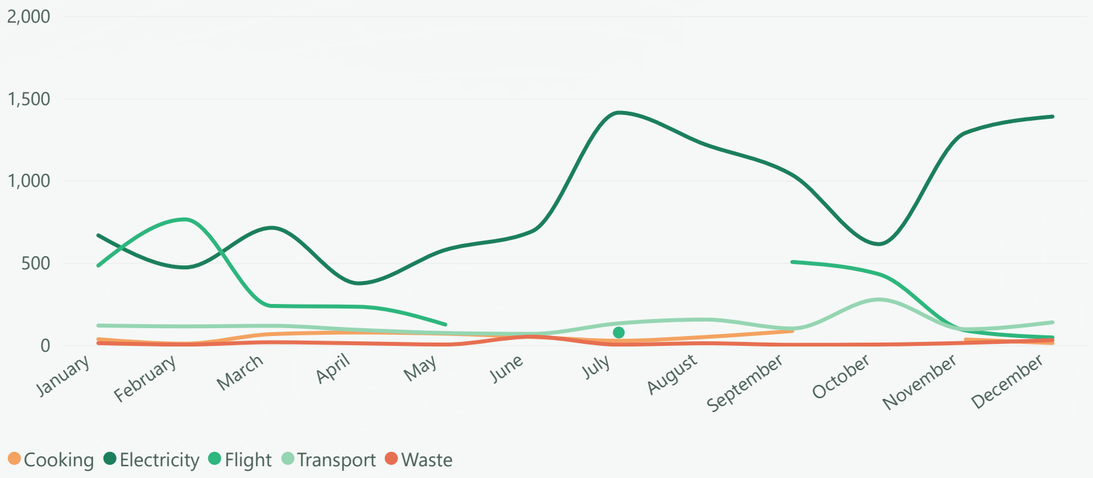
## Emission Share by Category

Total CO2 by Category




## Monthly $CO_2$ Trend by Category

Total CO2 by Month and Category




## Activity Quantity vs $CO_2$ Emissions

Sum of Quantity and Sum of CO2_Emission_kg by Category




Number of Activities

**180**



# SUSTAINABILITY ANALYSIS

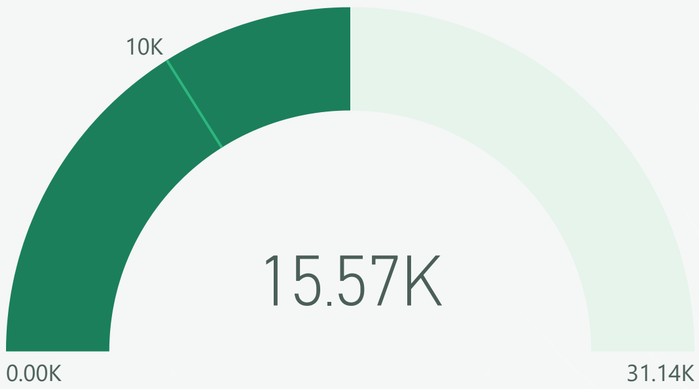
Progress against the $CO_2$ target and the biggest levers for reduction

Year

Year

## $CO_2$ Emissions vs Target

Total CO2 and CO2 Target




Total CO2

**15.57K**

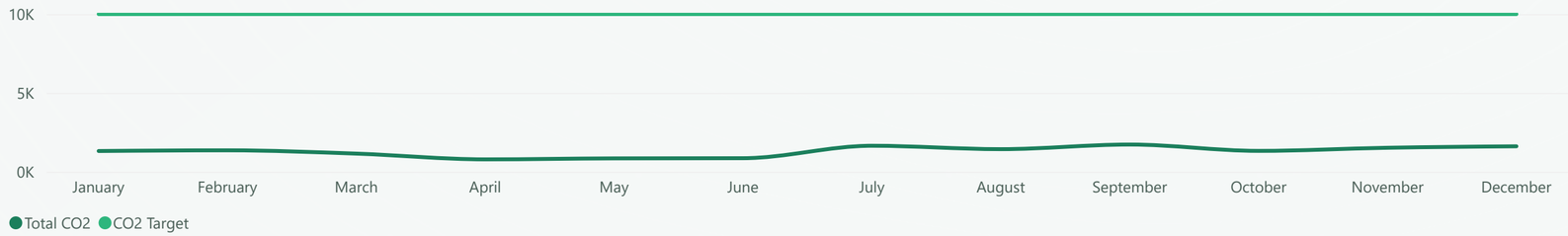
CO2 Target

**10K**



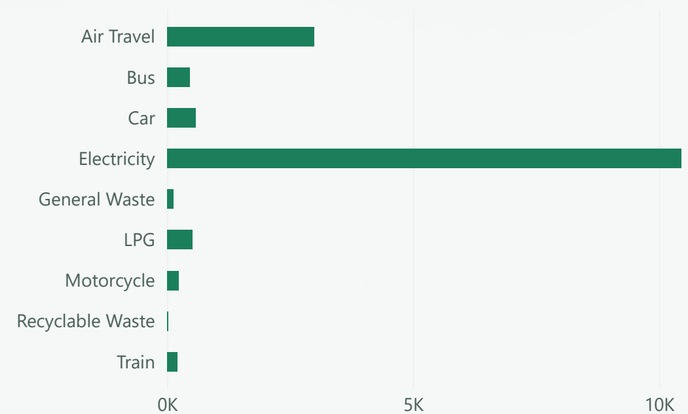
## Actual $CO_2$ vs Target by Month

Total CO2 and CO2 Target by Month



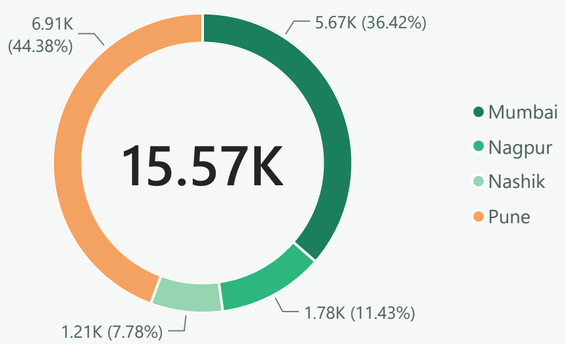

## Highest-Impact Activities

Total CO2 by Activity




## Emissions by Location

Total CO2 by Location






## Path to Net Zero

1. Focus first on the top-emitting activities
2. Shift energy use to renewables
3. Choose low-carbon transport options
4. Review progress monthly against target

Small changes, repeated every month, add up.

# ML INSIGHTS

Forecasting, anomaly detection and clustering on emission data | Python: Holt-Winters, Isolation Forest, K-Means



Forecast $CO_2$, next 6 months (kg)
**10.22K**

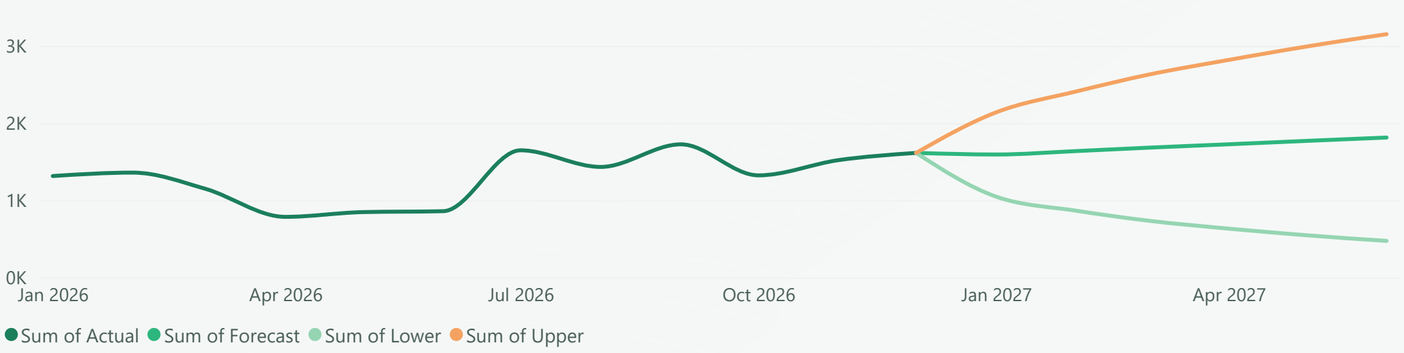
Anomalies flagged
**9**

High-impact activities
**2**

Records analysed
**180**

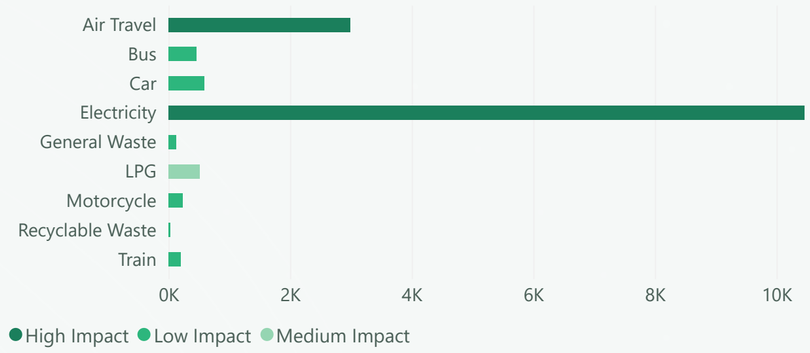
## Monthly $CO_2$: Actual vs 6-Month Forecast (with 95% band)

Sum of Actual, Sum of Forecast, Sum of Lower and Sum of Upper by Month




## Activity Impact Clusters (K-Means)

Sum of TotalCO2 by Activity and ImpactCluster




## Most Unusual Records (Isolation Forest)

| Date | Location | Category | Activity | Quantity | CO2_Emission_kg | AnomalyScore |
|---|---|---|---|---|---|---|
| Monday, September 28, 2026 | Mumbai | Flight | Air Travel | 1,748.00 | 262.20 | 0.66 |
| Monday, November 16, 2026 | Pune | Electricity | Electricity | 496.00 | 406.72 | 0.64 |
| Friday, July 03, 2026 | Pune | Electricity | Electricity | 487.00 | 399.34 | 0.62 |
| Wednesday, January 21, 2026 | Mumbai | Flight | Air Travel | 1,040.00 | 156.00 | 0.62 |
| Friday, October 16, 2026 | Mumbai | Transport | Car | 442.00 | 84.86 | 0.61 |
| Sunday, February 22, 2026 | Mumbai | Flight | Air Travel | 1,713.00 | 256.95 | 0.61 |
| Wednesday, February 18, 2026 | Pune | Flight | Air Travel | 1,695.00 | 254.25 | 0.61 |
| Saturday, October 24, 2026 | Mumbai | Flight | Air Travel | 1,391.00 | 208.65 | 0.61 |
| Thursday, April 30, 2026 | Pune | Transport | Motorcycle | 436.00 | 44.91 | 0.61 |



## How the models work

- Forecast: Holt-Winters smoothing projects monthly $CO_2$ six months ahead, with a 95% band.
- Anomalies: Isolation Forest flags the ~5% most unusual records.
- Clusters: K-Means groups activities into Low, Medium and High impact.

Note: indicative forecast, based on limited history.